Maritime detection: RF-DETR Full Fine Tuning

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What is RF-DETRs?

- **RF-DETR** (Roboflow, 2024) — DINOv2-backboned, NMS-free real-time detection transformer; current open SOTA on COCO and RF100-VL, ahead of YOLO11/YOLO26 at comparable latency.
- **Transformer-based detector**: DINOv2 backbone → multi-scale projector → lightweight 3-layer deformable-attention decoder → end-to-end set prediction (no anchors, no NMS post-processing step).
- **Small variant (~22M params)** — closest capacity match to YOLO11s among Apache-2.0-licensed sizes; a good fit for our ~2.5K image, 7-class dataset without the extra cost of Large/XL.
- Pretrained on **Objects-365 (pseudo-labeled with SAM2) + COCO** — broad, general-purpose detection knowledge before fine-tuning.
- Supports **direct fine-tuning on YOLO-format datasets** (dataset_file="yolo") — no dataset conversion needed Trained/evaluated at multiple resolutions, so accuracy/latency can be traded off at inference time without retraining.

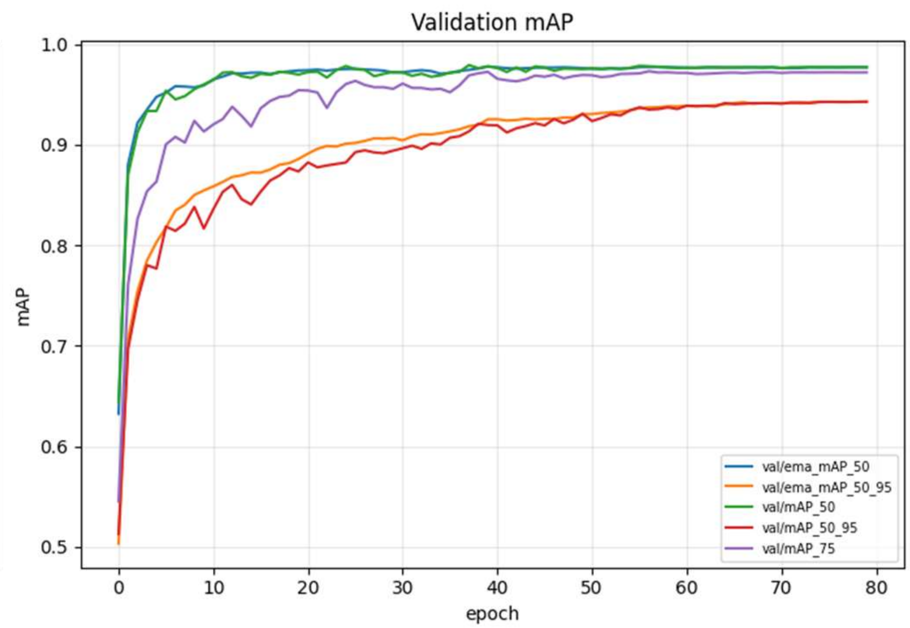
Fine-Tuning Strategy Used

•**Goal**: Adapt Objects-365/COCO-pretrained RF-DETR-Small to 7 marine classes (boat, buoy, cargo, ferry, tug, warship, yacht).

•**Strategy**: **Full fine-tuning** — no layer freezing, entire network (backbone + projector + decoder) trained end-to-end.

•**Why full fine-tune, not partial-freeze like YOLO**: RF-DETR's DINOv2 backbone already uses a **lower learning rate for the backbone** (lr_encoder) than the rest of the network (lr) by default — this achieves the same "protect general features, adapt task-specific ones" goal as YOLO's freeze=4, but built into the optimizer rather than via hard layer freezing.

Important curves



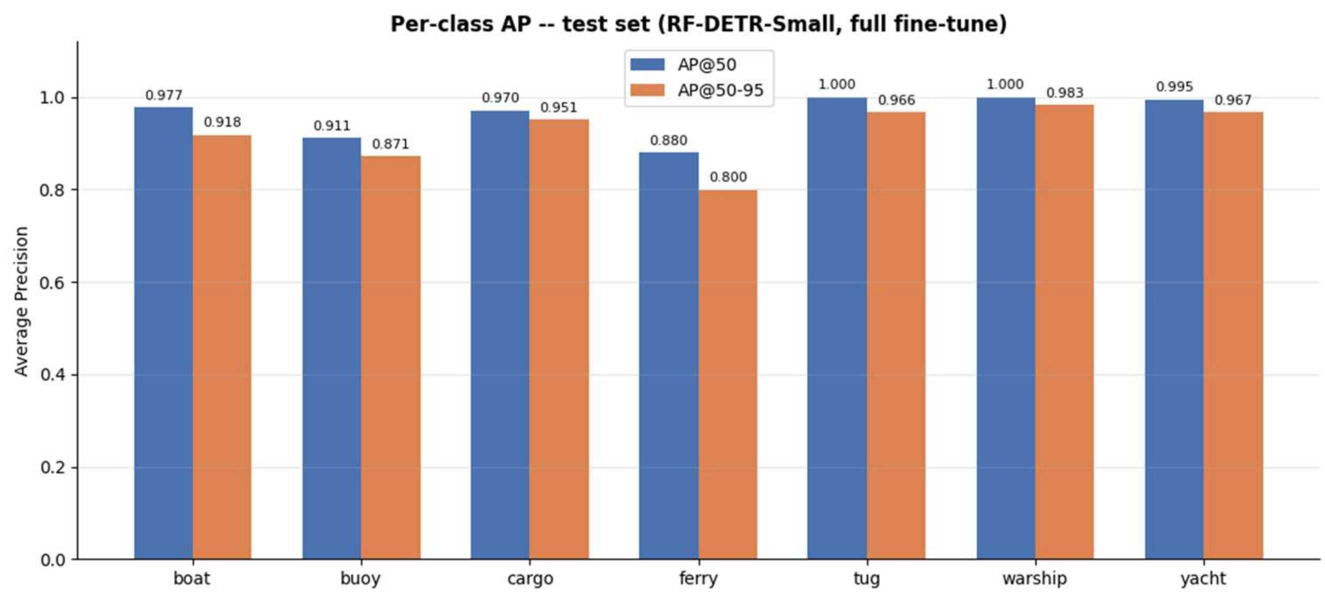
Overall test metrics

mAP50 : 0.96

mAP50-95 : 0.92

AR (avg recall, maxDets=100) : 0.93

Per class Evaluation metrics



Sample output images: Ground truth vs Predicted